BIJAN POURAZARI

ARTIFICIAL INTELLIGENCE RESEARCHER & SENIOR FULL-STACK SYSTEMS ENGINEER

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1. EXECUTIVE SUMMARY & AI SPECIALIZATION

Artificial Intelligence Specialist & Senior Full-Stack Engineer with over a decade of technical execution spanning foundational AI research, distributed systems architecture, and production software engineering. Academic research background rooted in early Symbolic AI, Constraint Satisfaction Problems (CSPs), state-space search algorithms, and Vector Calculus at the **Donald Bren School of Information and Computer Sciences (UC Irvine)** and Electrical Engineering (EECS). Extends core theoretical AI principles into modern production architectures—including AI provider query routing (OpenAI/Gemini) in precision agriculture (*agronomy-studio*), interactive LLM verification systems (*LLM-Workflow-Academy*), and resilient event-driven distributed platforms. Holds a Bachelor of Science in Computer Science from California State University (2021) and an Associate of Science in Computer Science from Saddleback College (2019).

2. FOUNDATIONAL AI & SYSTEMS RESEARCH (UC IRVINE — DONALD BREN SCHOOL / EECS)

Conducted academic research and rigorous foundational coursework in Artificial Intelligence, Constraint Satisfaction, Computer Architecture, and Multivariable Mathematics during the 2012–2013 academic term at UC Irvine's Center for Machine Learning and Intelligent Systems (CML) and Donald Bren Hall:

Research / Domain Area	Algorithmic Frameworks & Technical Implementation	Direct Mapping to Modern AI & Systems
Symbolic AI & Constraint Satisfaction (CSPs) COMPSCI 171 / 175 / 275 Prof. Rina Dechter Lab	Engineered deterministic search engines and Sudoku / n-Queens solvers utilizing **recursive backtracking algorithms**, constraint networks, and state-space search heuristics.	Foundational logic underpinning modern AI reasoning tree rollouts, exact search trees, and decision-path pruning.
Adversarial Search & Decision Theory COMPSCI 171 Center for Machine Learning	Implemented **Alpha-Beta Pruning** and game tree evaluation algorithms to optimize decision boundaries and eliminate suboptimal branches in adversarial state spaces.	Core tree-evaluation math utilized in modern Reinforcement Learning (RL) agents and MCTS-based LLM reasoning engines.
Vector & Multivariable Calculus MATH 2D / 2E Dept. of Mathematics	Mastered vector fields, partial derivatives, gradient vectors, line/surface integrals, and **Green's / Stokes' / Divergence Theorems**.	The exact mathematical mechanism behind **Backpropagation** in Deep Neural Networks (multidimensional gradient chain rule).
Computer Architecture & Parallel Hardware COMPSCI 151 / EECS 112 EECS Program	Analyzed ALU architecture, instruction-level parallelism, pipelining, CPU cache performance, and memory hierarchy constraints.	Essential low-level understanding for parallel GPU compute cluster optimization and high-throughput AI inference.

3. APPLIED ARTIFICIAL INTELLIGENCE & ENGINEERING PROJECTS

agronomy-studio — Precision Agriculture & NLU Integration — Blazor WebAssembly | TypeScript Microservices | OpenAI & Gemini API Provider Routing | GIS

• Developed an intelligent precision agriculture platform integrating CIMIS, NRCS SSURGO, Open-Meteo, and Google Maps APIs.
• Architected a natural-language query interface with dynamic multi-provider AI routing (OpenAI and Gemini) to translate complex soil, weather entropy, and irrigation datasets into actionable farm management recommendations.
• Implemented serverless Netlify Functions microservices maintaining strict domain boundaries per service module.

LLM-Workflow-Academy — Static AI Education & Verification Platform — Angular 21 | Signals | Markdown Engine | Quiz Verification | Local Storage Persistence

• Engineered a host-agnostic, zero-backend continuing education platform focusing on practical Large Language Model integration.
• Authored course modules covering executor-agnostic LLM workflows, probabilistic verification patterns, syntax highlighting, and Mermaid diagram rendering.
• Structured content delivery through versioned Git submodules combined with reactive Angular 21 state signals.

Academic AI Research Publications — Economics & Ethics of Automation — Bachelor's Degree Capstone Papers | CST300 | California State University

• *Impact of Automation on the Economy:* Authored an academic ethics paper evaluating the economic displacement caused by AI and autonomous systems, advocating for proactive workforce re-education and state-level technology adoption.
• *Computer Software Industry Analysis:* Conducted comprehensive research on software architecture trends, cloud computing, machine learning adoption, and technical leadership pathways.

4. FULL-STACK & DISTRIBUTED SYSTEMS INDUSTRY EXPERIENCE

Software Engineer — Premier Crop Systems (Nov 2020 – Present)
• Architect and maintain enterprise Angular UI features, C# ASP.NET Core Web APIs, SQL Server stored procedures, and Azure Blob Storage workflows.
• Engineer high-performance data access layers using Dapper and Entity Framework Core to process complex agricultural telemetry.
• **Business Impact:** Contributions directly supported **10% year-over-year revenue growth for four consecutive years**.

Independent Software Developer — Provenza Floors & Enterprise Clients (Jul 2013 – Oct 2020)
• Built and released native Android (Java) and iOS (Swift) mobile applications with dynamic JSON rendering from Google Cloud Storage.
• Modernized enterprise WebForms applications into ASP.NET MVC, Web API, and AngularJS architectures.
• Supported digital systems that enabled 10% YoY revenue growth across consecutive client contracts.

5. FEATURED CORE ENGINEERING REPOSITORIES

roguenet-live-services — *.NET 10 | Microservices | Idempotency | Transactional Outbox Pattern | Append-Only Ledger*
• Always-online gaming backend featuring idempotent reward processing, optimistic concurrency control, and outbox event publishing for guaranteed zero-loss message delivery.

mvp-finance — *.NET 10 Minimal APIs | EF Core | SQLite | Angular 21 Signals | Event Sourcing*
• Local-first personal finance platform featuring an event-sourced accounting engine for multi-account ledger tracking, spendable-now calculations, and reactive UI state.

GreeksVisualizer — *C# | TypeScript | Quantitative Financial Modeling | Risk Analytics*
• Options market risk visualization tool translating complex Black-Scholes differential calculus into intuitive, real-time risk metric charts.

6. EDUCATION & PUBLIC RECORDS VERIFICATION

Academic Degree & Record	Institution & Year	Focus & Highlights
Bachelor of Science (B.S.) in Computer Science	California State University (Graduated 2021)	Computer Science Core, Software Engineering, AI & Automation Ethics Capstone.
Associate of Science (A.S.) in Computer Science	Saddleback College (Graduated 2019)	Object-Oriented Programming, Data Structures, Algorithms, Discrete Math.
Foundational AI & EECS Coursework	UC Irvine — Donald Bren School / EECS (2012–2013)	Constraint Satisfaction (COMPSCI 171/175), Architecture (COMPSCI 151), Calculus (MATH 2D/2E).
Official Court Record Disposition Orange County Superior Courts Harbor Justice Center, Case #18-002738	California State Records Dispo Date: 09/16/2021	**Case Status: Fully Dismissed / Disposed** pursuant to Penal Code 1203.4. All historical charges completely exonerated/dismissed.